The Physical Limit of Operational Neural Hypoxia Detection in the Black Sea

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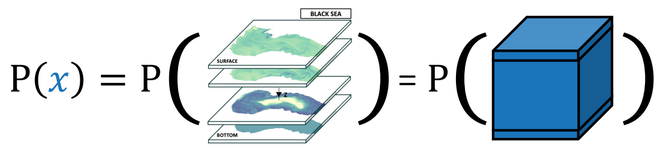
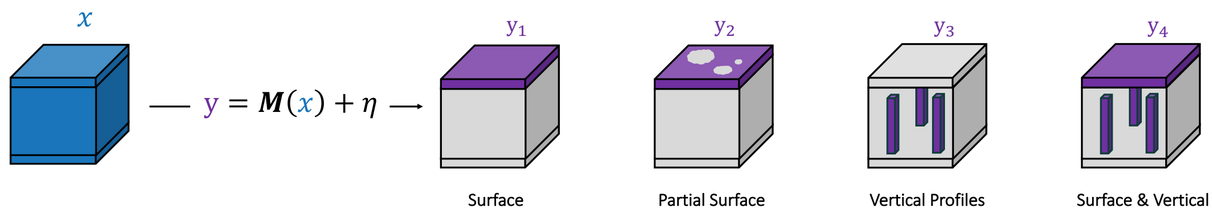
Can we accurately **estimate** ocean **oxygen** levels from Earth observations ? If so, can we reliably **detect hypoxia** ?

Problem Statement

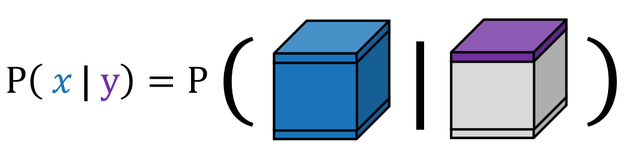
The Black Sea can exist in different states x, but some are more probable than others. We define a prior distribution over x using a physical-biogeochemical numerical model



An observation of the Black Sea, denoted as y, provides partial information about x



Given an observation y, we aim to solve the Bayesian inverse problem that links satellite surface observations to the corresponding sub-surface state



Results

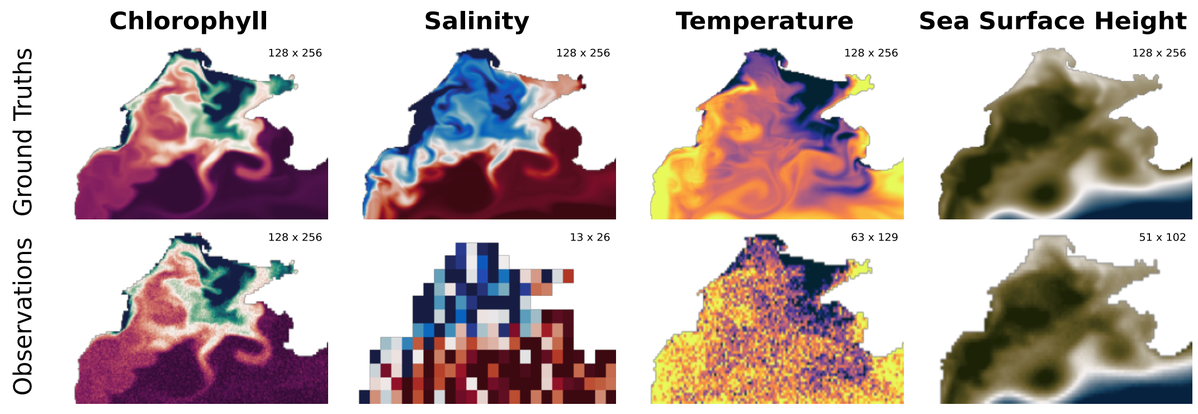


Figure 1: Visualization of Black Sea surface obsevations y generated using our realistic observation model $p(y \mid x)$ for an arbitraty state x in our test dataset. Using this observation model, we generate y at realistic satellite resolution and explicitely model instrument errors by adding a bias and random noise to the final observation.

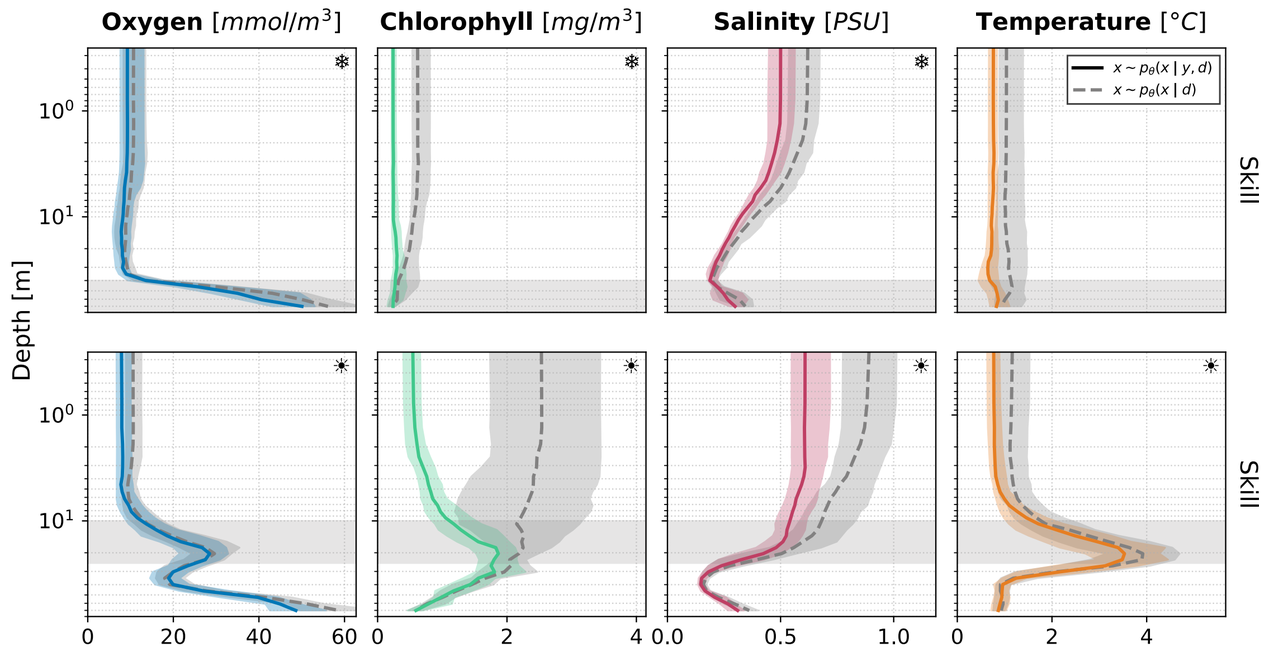


Figure 2: Skill (bias between ensemble mean and ground truth, lower is better) median and quantiles (25–75%) for ensembles x generated with (colored) and without (grey) surface observations y in winter and summer. Surface observations reduce estimation error for all emulated variables, including unobserved oxygen, with the largest improvement for oxygen in summer. These improvements are confined to the mixing layer, down to its maximum depth (grey region), consistent with the physical expectation that surface conditions constrain only the well-mixed upper ocean.

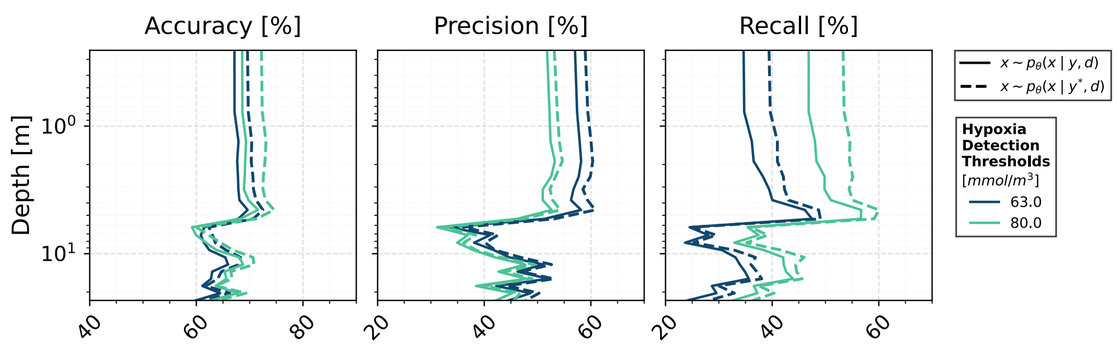
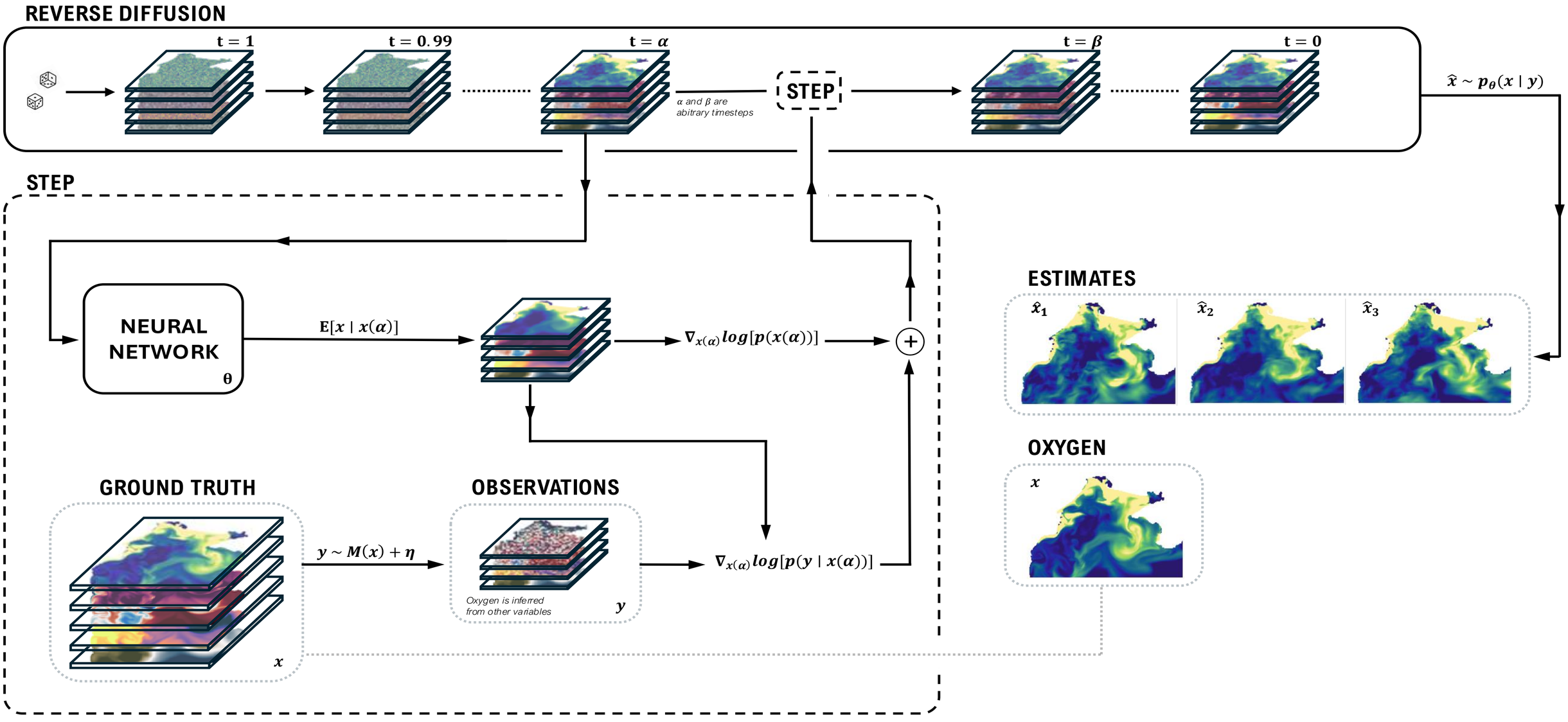


Figure 3: Mean classification metrics as a function of depth, averaged over June-September of two test years. Results are shown for two hypoxia detection thresholds 63 and 80 $[\mu mol\,.\,l^{-1}]$ and two observations type: realistic observations y (solid lines) and high-resolutions y^* (dashed). With the highest detection threshold, realistic observations achieves on average over the whole depth range (0-25 $[m]$), 60 [%] accuracy, 47 [%] precision and 45 [%] recall. Switching to high resolution observations improves all metrics, but only moderately, suggesting that surface observations alone are insufficient to accurately estimate oxygen levels.

Methodology